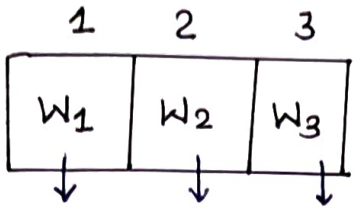
Day 1

1). factorial →



1 posn '3' choices
2 posn '2' choices
3 posn '1' choice

total nos of ways of arrangement
⇓
(order matters)
⇓
$W_1 \times W_2 \times W_3$
$3 \times 2 \times 1 \Rightarrow 6$ or $3!$

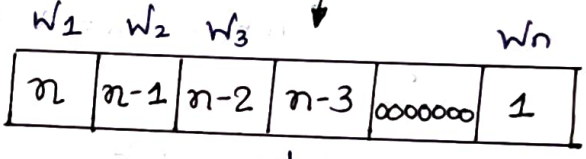
3 letters (A, B, C)
↓
change relative position
↓

different → A B C, A C B, B A C, B C A, C A B, C B A.

same

"N" different items or object taken all at a time

without replacement / order matters



total nos of ways
⇓
$W_1 \times W_2 \times W_3 \times \cdots \times W_n$
(fundamental principle of counting)
↓
$(n)(n-1)(n-2) \cdots (1)$
↓
$\boxed{n!}$

problems :

① How many n-digit nos can be formed using 1, 2, 3, 7, 9 w/o any repetition of digits when $n=5$?

↑ distinct object

5 diffo item taken all at a time

move → $\underline{5}$, $\underline{4}$, $\underline{3}$, $\underline{2}$, $\underline{1}$

fixed → P_1 P_2 P_3 P_4 P_5

$\Rightarrow 5!$

② How many 4 digit nos can be formed using 0, 1, 2, 3, 4, 5 ?

$\underset{\downarrow}{5}$ (1,2,3,4,5), $\underset{\downarrow}{5}$ (0,1,2,3,4,5), $\underline{4}$, $\underline{3}$ $\Rightarrow$ 300 ways.

2). permutation :-

'N' item taken 'r' at a time

A, B, C (3-letter)

↓

taken two at a time

AB AC BC
BA CA CB

6-cases

} different arrangement (or matters)

general

w_1	w_2	w_3	...	w_n
n	$n-1$	$n-2$	...	$n-(r-1)$

total nos of ways (total arrangement)

↓

$(n)(n-1)(n-2) \dots (n-(r-1))$

↓

$$\frac{(n)(n-1)(n-2)\dots(n-(r-1))(n-r)!}{(n-r)!}$$

⇓

$$^nP_r \Rightarrow \frac{n!}{(n-r)!}$$

'r' different item taken 'r' at a time. [Order-matters].

★ 'n' different items taken all at a time [repetition-allowed].

$\Rightarrow n \times n \times n \dots$ N times

$\Rightarrow n^N$ ways.

AAA ABA ACA BAA
AAB ABB ACB BAB ....
AAC ABC ACC BAC

3). Combination :- 'n' different items, 'r' selected at a time, where order doesn't matter

'n' different items → group / selection
→ order doesn't matter (W/O repetition)

A, B, C → taken two at a time

AB, BA → same
BC, CB → same
CA, AC → same

↳ arrangement ⇒ $^3P_2$ → repetition removed → $\frac{^3P_2}{2!}$

* total nos of selection $\Rightarrow \frac{^nP_r}{r!} \rightarrow {}^nC_r \rightarrow \frac{n!}{(n-r)!\,r!}$

Combination with repetition :

W/O repetition
AB
AC
BC

With repetition
AA, AB, AC, BB, BC, CC → we also get AA, BB, CC

using the formulae.

$^{n+r-1}C_r$

Sum up-

	No repetition	Repetition Allowed
order matters	^nP_r	n^r
order doesn't matters	^nC_r	$^{n+r-1}C_r$

Q) There are (5 letters) & (4 boxes). Then in how many ways we can post these letters?

Not a permutation problem → Because you aren't arranging the letters; you are just assigning each letter to one of the boxes

Not a combination problem either, because you aren't choosing a subset of letters or boxes

5 letters, 4 letter boxes

for each letter, you can choose any one of the 4 boxes

move	L1	L2	L3	L4	L5
	↓	↓	↓	↓	↓
fixed	4 ways	4 ways	4	4	4

has 4 possible boxes, independently of where first letter was placed

⇒ 4^5 ways ⇒ 1024 ways

suppose you have 5 distinct letters & 4 letterboxes

① How many ways can you place exactly 3 letters in different letterboxes?

choosing 3 letter from 5 and each letter must go into a different letterbox

$^5C_3 * 3!$ → 3 diff letter, 3 diff boxes to assign $= {}^nP_r$

10 × 6 ⇒ 60 ways

(selection of 'r' diff. objects out of n-diff. objects) * (arrangement of 'r' objects) ⇒ (arrangement of n-diff. objects taken 'r' at a time)

Q) In how many ways (5 persons) can be seated on (8 chairs)?

5 persons → move

8 chairs → fixed

move $P_1 \times P_2 \times P_3 \times P_4 \times P_5$

↓ ↓ ↓ ↓ ↓

8ways 7ways 6 5 4

repetition not allowed

↓

only one person sit at a time

↓

All the person want to be seated

Q) There are (3 rings) and (4 fingers) then in how many ways we can worn these rings if –

(3 rings) → move

(4 fingers) → fixed

(a) there is no restriction (repetition allowed)

$R_1 \times R_2 \times R_3$

↓ ↓ ↓

4ways 4ways 4ways → 64 ways

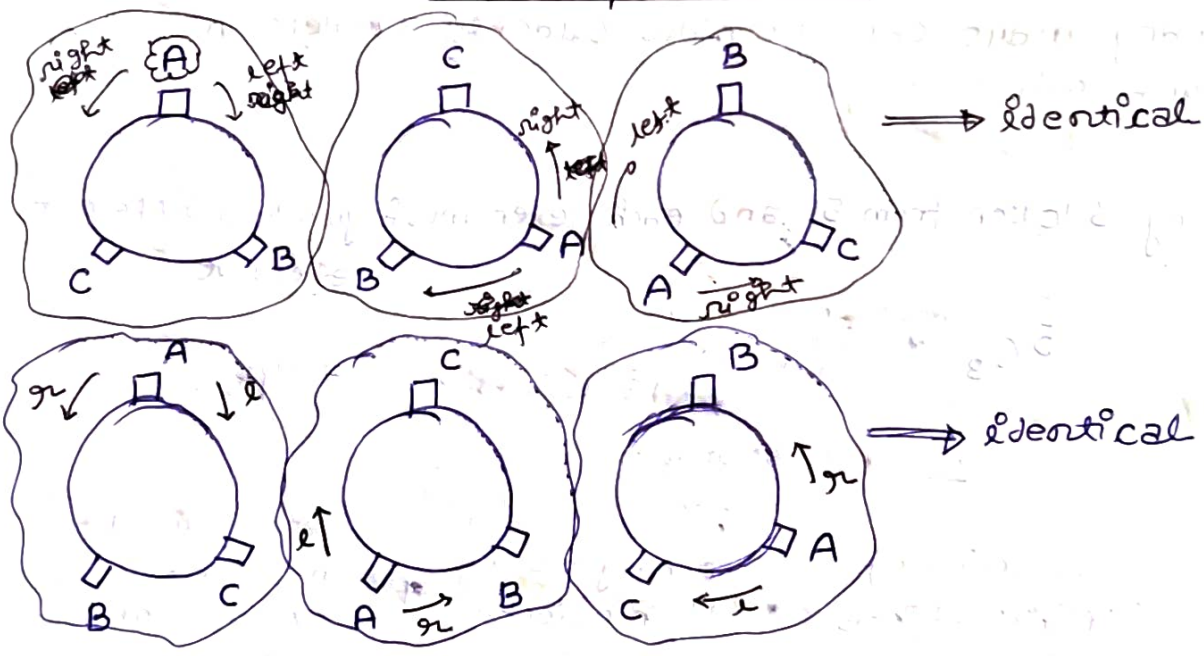
(b) with atmost one ring in a single finger (repetition not allowed)

$R_1 \times R_2 \times R_3$

↓ ↓ ↓

4 3 2 → 24 ways

circular permutation



in linear → in circle

3 arrangement → 1 arrangement count

$1 \longrightarrow \frac{1}{3} \times 3! \rightarrow$ placing

In linear placing/arranging 3 letter taken all at atime → 3!

In circular → $\frac{3!}{3} \Rightarrow 2$

If we have 'n' different objects
the nos of circular permutations will be –

$\frac{n!}{n} \rightarrow (n-1)!$

(when anti-clockwise & clockwise arrangments are different)

⇓ ie!

Q) In how many ways 4 boys can be arranged on a circular table?

→ (3!) Ans

$\frac{(n-1)!}{2}$

(when anticlockwise & clockwise arrangement are same)

⇓

Q) How many garlands can be made using 4 different flowers?



$\Rightarrow \frac{(4-1)!}{2} \Rightarrow \frac{3!}{2} \Rightarrow 3.$

→ Alike objects

ie MISSIPPI

p alike → SS

q alike → III

r alike → PP

M

nos of arrangement

↓

taken all at a time

$\frac{(p+q+r)!}{p!\,q!\,r!}$

$\Rightarrow \frac{(2+3+2+1)!}{2!\,3!\,2!}$

Q) How many words can be formed using all the letters of the word → "HONOLULU"

① without any restriction

H, N, O, O, L, L, U, U

$$\frac{8!}{2!\,2!\,2!}$$

② such that alikes of same kind are together

H, N, [OO], [LL], [UU]

$1 \times 1 \times 1 \times 5!$

Combination of Alike Objects

Consider some objects-

[A, A], [B, B, B], [C, C]

- How many ways we can select one object = 3 ways

→ to select one object from given set of objects

[A, A], [B, B, B], [C, C]
↓ A, ↓ B, ↓ C

- How many ways we can select two objects ⇒

→ both different → ${}^3C_2$

\+ ⇒ 6 ways

→ both alike → ${}^3C_1$

selecting 1 set of object from 3 set of object
every set give 2 objects

- How many ways we can select three objects ⇒ 3 ways

→ 3 different → ${}^3C_3$

\+

→ 1 diff, 2 alike → ${}^3C_1 \times {}^2C_1$ → 3 में से 1 select किया set for 1 diff; फिर 2 में से for 2 alike

\+

→ 3 alike → 1 → [B, B, B] only 1 way

Q) In how many ways we can select 4 letters out of the letters of the word → "BHANNAAT"

T, B, H, A, A, A, N, N

case I → 4 diff letters → $^5C_4$

case 2 → 3 alike, 1 diff → $^4C_1 \times 1$ → A, A, A

case 3 → 2 diff, 2 alike → $^2C_1 \times {}^{④}C_2$ → if AA choosed then T, B, H, N, N choices
→ if NN choosed then T, B, H, AAA choices

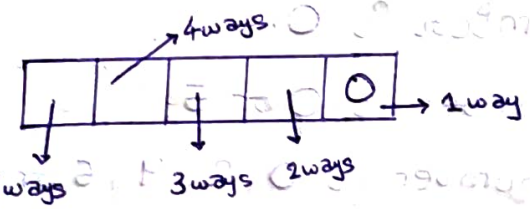
AAA
NN

case 4 → 2 alike, 2 other alike.
→ $^{②}C_2$ → 2 set of alike A, A, A N, N
दोनो ही लेने है।

Q) How many 5-digit numbers divisible by 2 can be formed using digits 0, 1, 2, 3, 4, 5 without repetition of digits?

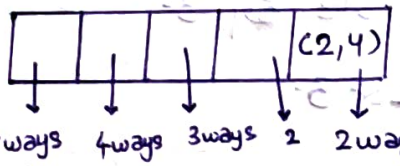
divisible by 2 → end with
0 | (2, 4)
case(A) | case(B)

case A)



5! → 120 ways

case B)



4 × 4 × 3 × 2 × 2 → 192 ways

total ⇒ 120 + 192 ⇒ 312 ways

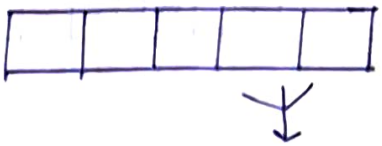
⊗, 1, ⊗, 3, ⊗, 5
↓
either of one is filled.

Q. How many five digits number divisible by 4 can be formed using digits 0, 1, 2, 3, 4?

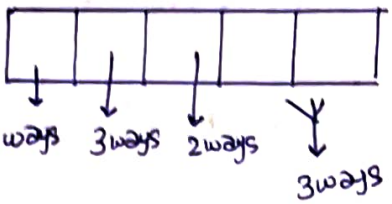


{divisible by 4}

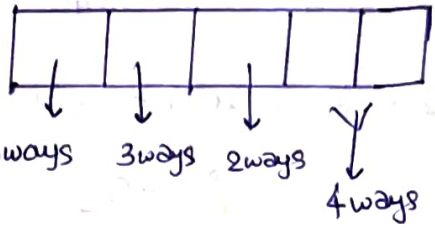
0 , 4
or
4 , 0
or
2 , 0



1 , 2 | 2 , 4
5 , 2 | 3 , 2



$4 \times 3 \times 2 \times 3$
$\Rightarrow 72$



$3 \times 3 \times 2 \times 4$
$\Rightarrow 72$ ways

$\Rightarrow$ total = 144 ways

→ Test of divisibility:

10 → if unit digit of number is 0.

5 → if unit digit of number is 0 or 5.

2 → if unit digit of number is 0, 2, 4, 6 or 8

3 → if sum of digit is divisible by 3.

6 → if divisible by ~~both~~ 2 & 3.

4 → if last two digits are divisible by 4

7 → ie num = 749

$74(9) \rightarrow 9 \times 2 = 18$ —(i)

$74 - 18 = 56$ —(ii)

56 is divisible by 7,

Hence 749 also.

Q. find the number of different words which can be formed using all the letters of the word "HISTORY"?

HISTORY

n' different item ⟶ taken all at a time

Total nos of Letters ⇒ 7!

Q. In how many ways can be Letter of the words "DELHI" be arranged so that the vowels occupy only even places?

DELHI → vowel : E, I

→ consonant : D, H, L



P1 P2 P3 P4 P5

odd even odd even odd

E I ? → diff (order matter)

I E

E E

I I

? → W1 for → 'n' diff. item taken all at a time

D H L ?

H D L

H L D → W2 for →

D H L

L D H

L H D

W1 × W2 ⇒ 3! × 2!

⇒ 12 ways

Q) In how many ~~ways~~ of these words do the vowels & consonants occupy the same relative posn as in - "COMBINE".

C O M B I N E
↓ ↓ ↓ ↓ ↓ ↓ ↓
cons vow cons cons vow cons vow : same relative posn.

C	V	C	C	V	C	V
①	②	③	④	⑤	⑥	⑦

V_1 V_2 V_3 } 3! ways
② ⑥ ⑦

C_1 C_2 C_3 C_4 } $C_1C_2C_3C_4$, $C_2C_1C_3C_4$, $C_2C_3C_4C_4$, $C_3C_2C_4C_4$
① ③ ④ ⑥
4! ways

total ways ↓
$W_1 \times W_2 \Rightarrow 3! \times 4!$
$\Rightarrow 144$ ways

Q) How many words can be formed using letters of the word "EQUATION"?

@ taken all at a time → 8! ways

@ how many of these begin with E and ends with N?

E Q U A T I O N
↓ ↓
fixed fixed

E	6	5	4	3	2	1	N
1 way							1 way

$\Rightarrow 1 \times 6!$ ways

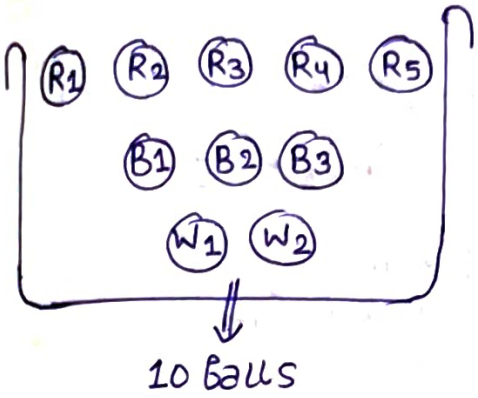
© How many of these end & begin with Consonant?

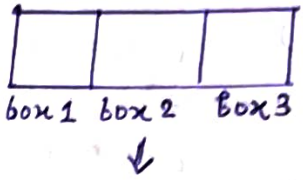
	6	5	4	3	2	1	

→ 2 ways (consonants)
← 3 ways (consonant)

Vowel: E, Q, A, O, I
Conson: T, Q, N

Q. In how many ways 5 different red balls, 3 diff. black balls & 2 diff. white balls can be placed in 3 diff. boxes. such that each box contain only one ball?





$^{10}P_3 \rightarrow \frac{10!}{7!} \Rightarrow 720$ ways